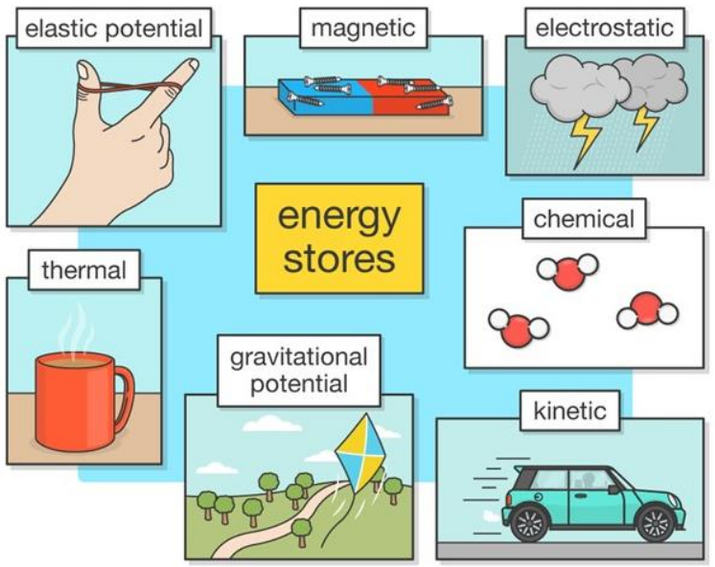
Knowledge Organiser
Year 7: Energy – Stores and Transfers

What is a store of energy?
- Energy can exist in different form that we call **stores of energy**
- All energy stores are measured using the unit **joules, J**
- Energy cannot be **created or destroyed,** it just moves between different stores

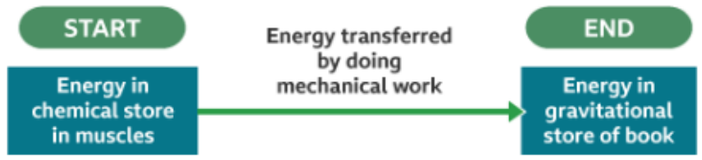
The forms that energy can be stored as:



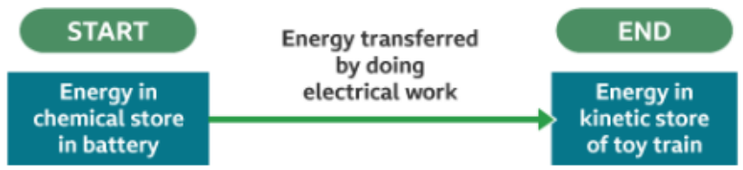
How can energy be transferred?
- Energy can be transferred between stores
- This can happen in four ways:

1. **mechanical** - force applied to move an object
2. **electrical** – when charge flows (electricity)
3. **heating** – transferred between hotter and colder regions
4. **radiation** –transferred as a wave, for example as light or sound

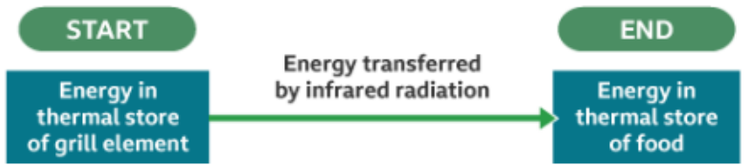
An energy transfer when a book is lifted onto a high shelf:



An energy transfer when a battery powered toy train is switched on:



An energy transfer when an electric toaster heats up bread:



How is energy wasted?
- When an energy transfer happens some of the **output energy** is useful and some is wasted
- Wasted energy is in a form that we do not want or need
- No energy transfers are perfect, some energy will always be wasted
- The energy will go into the surroundings

A table showing the input, useful output and wasted output energies in a range of energy transfers:

Energy transfer	Input	Useful output	Wasted output
A falling skydiver	Gravitational potential	Kinetic	Thermal, Sound waves
A boiling kettle	Electrostatic	Thermal	Kinetic, sound waves
TV	Electrostatic	Light and sound waves	Thermal
Radio	Electrostatic	Sound waves	Thermal
Light bulb	Electrostatic	Light waves	Thermal
Running jogger	Chemical	Kinetic	Thermal, sound waves
Squashing a spring	Kinetic	Elastic potential	Thermal, sound waves

What is efficiency?
- Efficiency is a measure of how usefully a device transfers energy
- If a device is 60% efficient, that means 40% of the energy is transfers is transferred to a wasteful store
- Most devices waste energy to the thermal store, think about how hot a light bulb gets!
- An efficient device will transfer most of its energy into a useful store

The equation for calculating percentage efficiency:

$$\text{efficiency} = \frac{\text{useful energy out}}{\text{total energy in}}$$

Example question: The energy supplied to a light bulb is 200 J. A total of 28 J of this is usefully transferred. How efficient is the light bulb?

$$efficiency = \frac{28}{200}$$

$$efficiency = 0.14$$

$$percentage\ efficiency = efficiency \times 100$$

$$percentage\ efficiency = 0.14 \times 100$$

$$percentage\ efficiency = 14\%$$

RAG Yourself!

Learning Outcomes;	RAG
Name the different forms of energy stores	
State the ways energy can be transferred	
Draw energy transfer diagrams	
Identify useful and wasted energy	
Calculate percentage efficiency	
Describe the differences between efficient and inefficient devices	

Knowledge Organiser
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The forms that energy can be stored as:

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